



CS776: Deep Learning for Computer Vision

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Spring 2026, CSE IIT Kanpur



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Agenda



Course Logistics



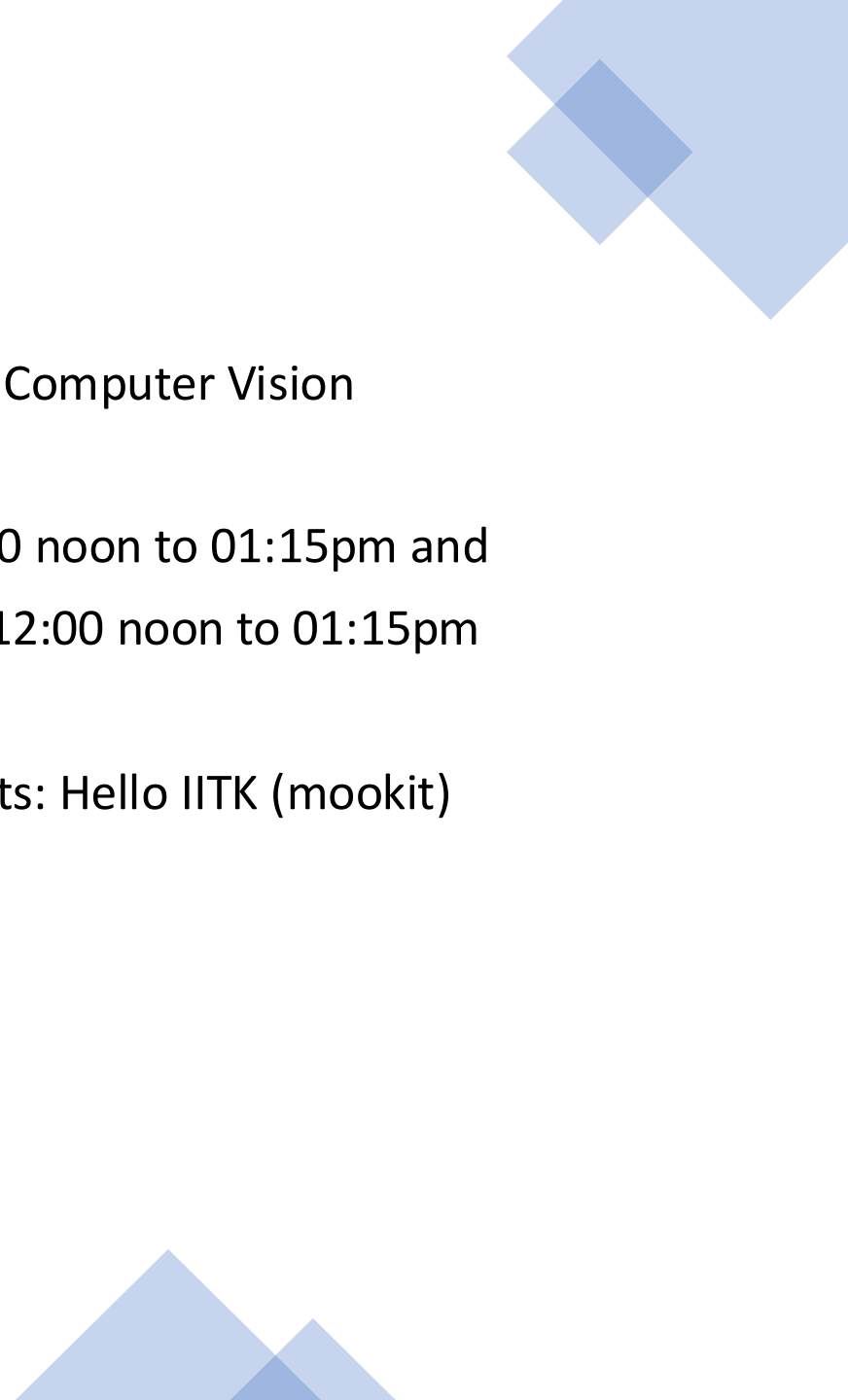
Discuss first
course handout



Introduction to
Computer Vision



Course Logistics

- CS776: Deep Learning for Computer Vision
 - Spring 2026
 - Class hours: Tuesday 12:00 noon to 01:15pm and
Wednesday 12:00 noon to 01:15pm
 - Classroom: RM101
 - Lecture slides, assignments: Hello IITK (mookit)
 - Quizzes/exams - in-class
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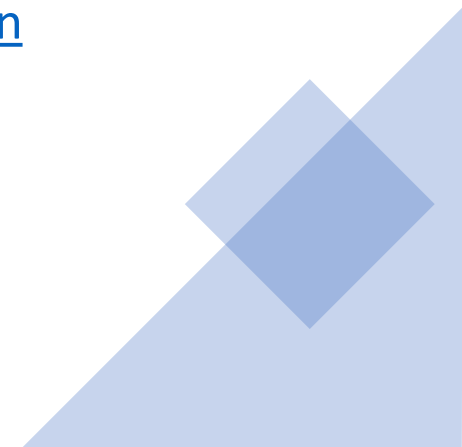


Contact Details: Instructor

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 - Important note: Please include CS776 in the subject to get the response.
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How to reach TAs and Instructor with doubts/questions?



TAs and I will be more than happy to answer your reasonable questions.



Please use Forums on Hello IITK for questions. It will help others with the similar questions too.



Email: Include Instructor and TAs in the email.

Please include "CS776" in the subject. I have filters setup in the email. If your email does not go in the folder, it will be ignored.

Course Prerequisites

- Strong understanding of basic probability, calculus and linear algebra is required.
- The students will be expected to have good programming skills.



Course Content

S. No	Broad Title	Topics
1	Introduction to Computer Vision	Applications, CV pipeline, image preprocessing, feature extraction
2	Deep Learning models	Concept of perceptron, activation functions, feedforward process, backpropagation, optimization algorithms
3	Convolutional Neural Networks (CNN)	CNN architecture, basic components of CNN< Image classification with CNN, optimizations
4	Model performance evaluation	Performance metrics, model evaluation, improving model, hyperparameter tuning, optimizing algorithms
5	Advanced models in CNN for classification and detection	LeNet-5, AlexNet, VGGNet, ResNet, R-CNN, SSD, YOLO
6	Transfer Learning	Basics of transfer learning, approaches, open source datasets
7	Generative adversarial networks (GAN)	GAN architecture, model evaluation

Books/References

- Deep Learning, Ian Goodfellow, Yoshua Bengio, and Aaron Courville
- Deep learning for Vision Systems by Mohamed Elgendy, Manning publisher, 2020 Edition
- Research papers from top-tier conferences and journals.

Grading Scheme

- Attendance (5%) - Acadly
- Quiz (20%)
- Assignment (10%)
- Mid semester exam (25%)
- Course project (40%)
- This is a tentative allocation.
 - Might change depending on the strength of the class and class progress
 - Relative grading
- Academic Integrity
 - Any type of plagiarism/cheating won't be accepted.

Questions?

